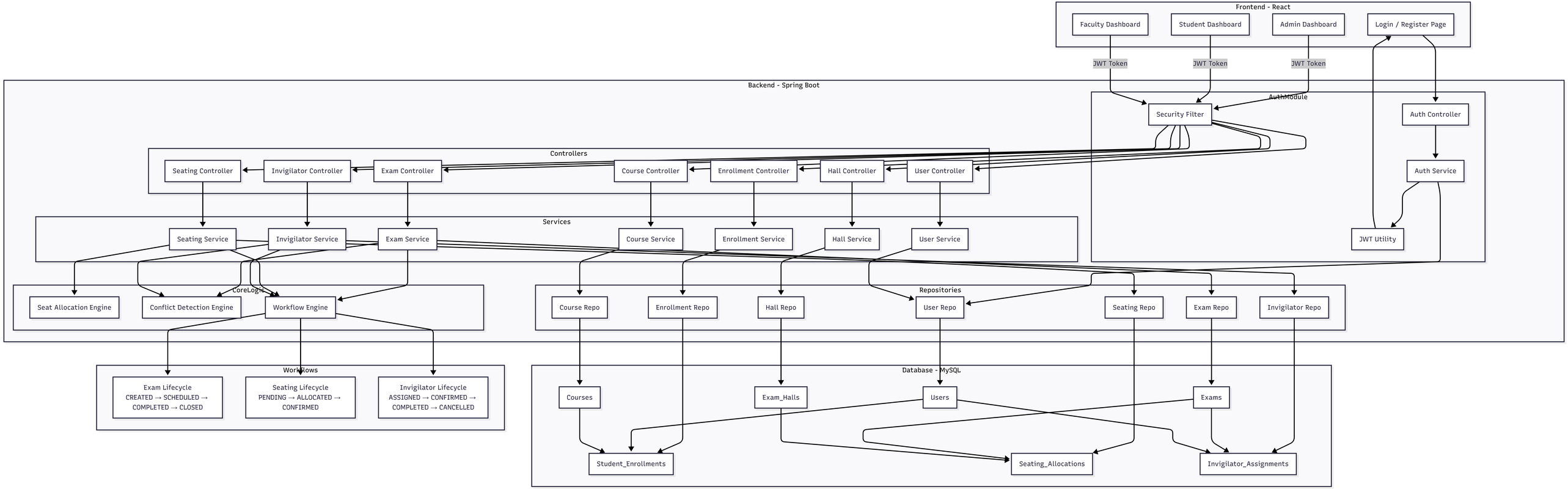
Frontend - React
Faculty Dashboard
Student Dashboard
Admin Dashboard
Login / Register Page
JWT Token
JWT Token
JWT Token
Backend - Spring Boot
AuthModule
Security Filter
Auth Controller
Auth Service
JWT Utility
Controllers
Seating Controller
Invigilator Controller
Exam Controller
Course Controller
Enrollment Controller
Hall Controller
User Controller
Services
Seating Service
Invigilator Service
Exam Service
Course Service
Enrollment Service
Hall Service
User Service
CoreLogic
Seat Allocation Engine
Conflict Detection Engine
Workflow Engine
Repositories
Course Repo
Enrollment Repo
Hall Repo
User Repo
Seating Repo
Exam Repo
Invigilator Repo
Workflows
Exam Lifecycle
CREATED → SCHEDULED → COMPLETED → CLOSED
Seating Lifecycle
PENDING → ALLOCATED → CONFIRMED
Invigilator Lifecycle
ASSIGNED → CONFIRMED → COMPLETED → CANCELLED
Database - MySQL
Courses
Exam_Halls
Users
Exams
Student_Enrollments
Seating_Allocations
Invigilator_Assignments